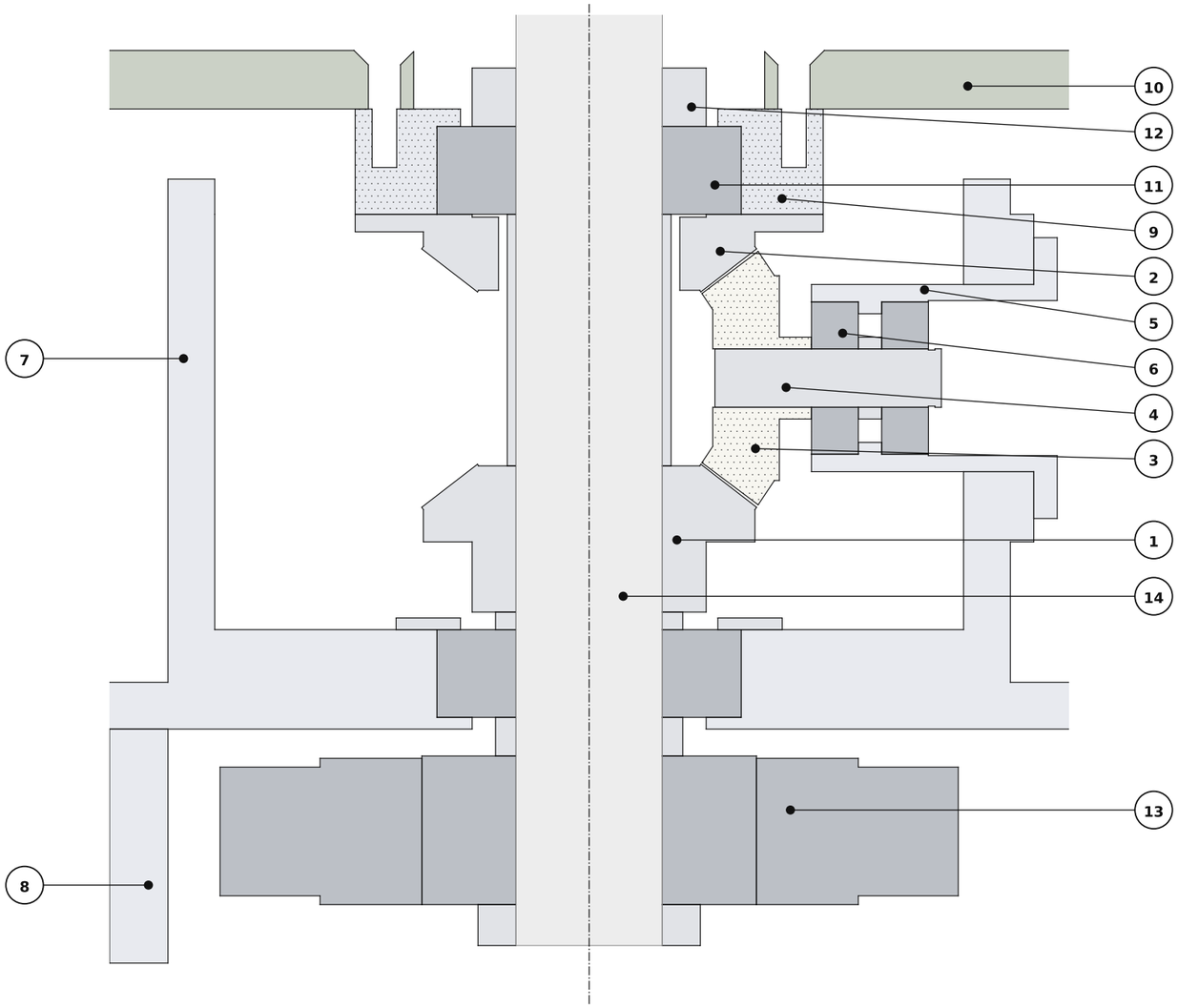
SECTION A–A: right through a pinion's axis (30°), left at 210° 1:1



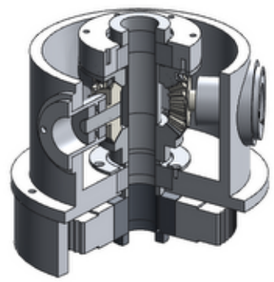
REVISIONS		
Rev	Date	Description
A	2026-10-11	First issue, draft for quotation

PARTS LIST

item	drawing	description	material	qty
1	DRV-101	side gear, the rotor's	1.4305	1
2	DRV-102	side gear, the counter-rotor's	1.4305	1
3	DRV-103	pinion z 20	POM-C	3
4	DRV-104	stub shaft	1.4305	3
5	DRV-105	cartridge sleeve	Al 6082	3
6	P-012	bearing SKF 6000-2Z	—	6
7	DRV-106	gear case	Al 6082	1
8	DRV-107	bell	Al 6082	1
9	DRV-108	counter-rotor hub	Al 6082	1
10	DRV-109	end disc	G10	1
11	P-001	bearing SKF 6205-2Z (of P-001's)	—	2
12	DRV-190	collar, spacers, key, ring, nut	1.4305	10
13	P-014	pulley 80T with its bush	—	1 + 1
14	SHF-301	shaft, its drive end (for reference)	—	ref.

NOTES

1. Status: the first cut of docs/drive-gear-belt.md on the record's 940 mm stack-up (z from the record's bottom end-bearing slot); the ledger asks for the drive to be re-cut on the layout (§6, item 71). PROPOSED.
2. Three POM pinions at 30° / 150° / 270° between two equal stainless side gears; the case is the carrier, held by the frame: the counter-rotor turns at −600 rpm when the rotor turns at +600 (§5). The teeth clear by 0.043 mm at the closest point (sim/unmodelled_parts_results.json).
3. On the shaft, clamped by the lock nut against the collar: the gear-end bearing's inner ring, spacer A, the rotor gear (keyed), spacer B, the locating bearing's inner ring, spacer C, the pulley's bush. Both bodies' weight (689 N) reaches the case floor's lip through the locating bearing (§7).
4. Departures from the first cut to make it buildable [IR]: the hub's lip above its bearing (so it carries the counter-rotor's 350 N); the counter-rotor's gear bolted by a flange outside the bearing; the pinions' bearings 12 mm apart, the mesh 17 mm outboard (the case's arithmetic; the statics' 16 / 12 do not fit it): 195 N at the limit, static safety 10; the bell Ø144 / Ø164, a C open to the motor, clear of the belt.
5. Backlash j_n 0.10-0.15 by shims under the cartridges' flanges and the rotor gear's spacer; POM-compatible grease in the closed case (§5).



3-D view, a quarter cut away; the end disc and the shaft left out (axonometric, not to scale)

DCCREG turbine · the drive

The 1 : −1 bevel reverser, side A

Drawing no. **DCCREG-DRV-100**

Material:

Mass: —	Qty per machine: 1	Rev A Sheet 72 of 103 A3
Scale: 1:1	Tolerances: ISO 2768-mK	Date: 2026-10-11
Drawn: generated by docs/make_drawing_set.py	Checked: — Approved: —	**PROPOSED · DRAFT**